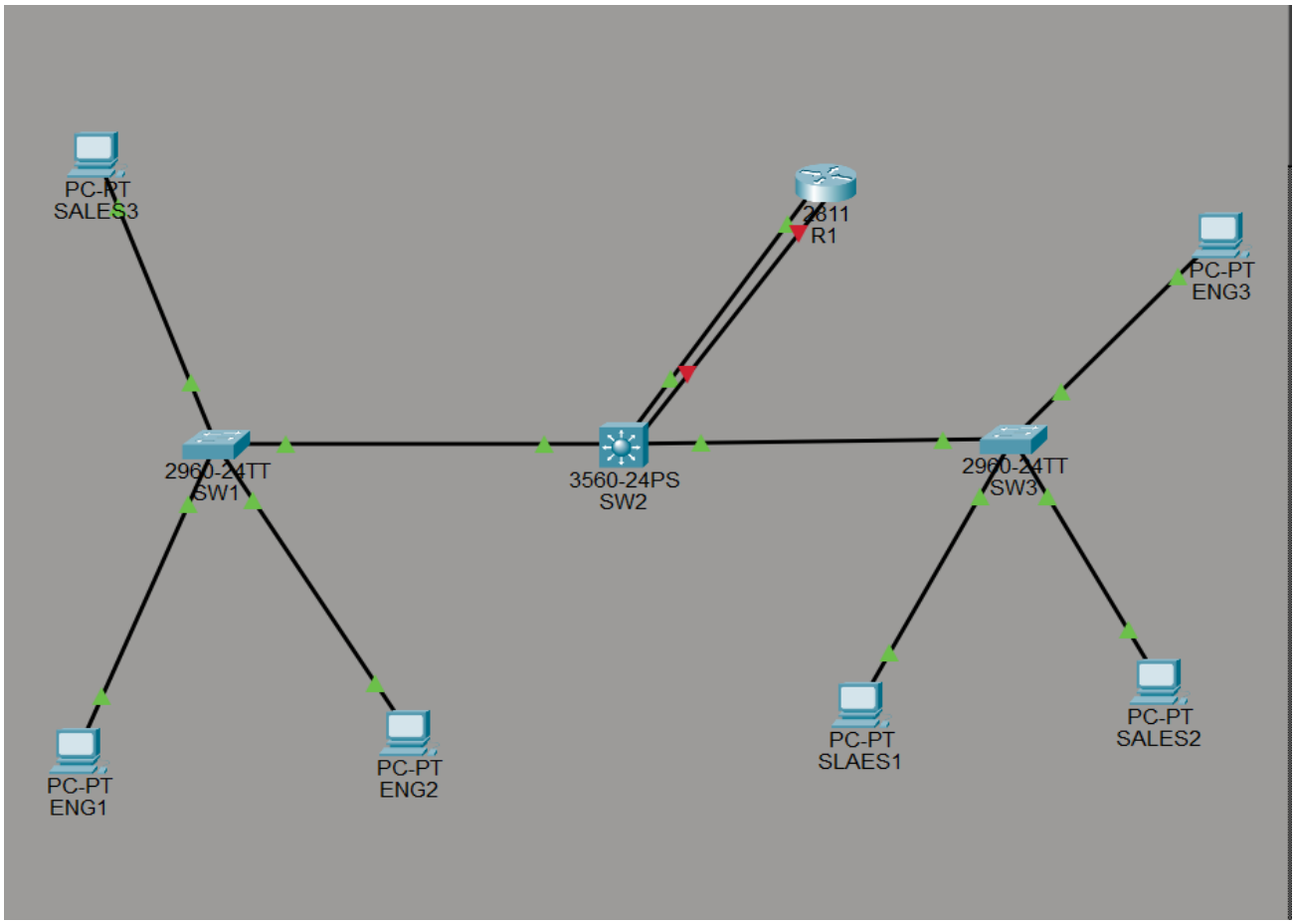
Week 9 VLAN and Inter-VLAN Routing Configuration



SW2

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
Switch(config)#vtp mode server
Device mode already VTP SERVER.
Switch(config)#vtp password cisco
Setting device VLAN database password to cisco
Switch(config)#
Switch(config)#end
Switch#write%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
CAMPUS.

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

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SW1

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
Switch(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch(config)#vtp password cisco
Setting device VLAN database password to cisco
Switch(config)#
Switch(config)#end
Switch#write%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
CAMPUS.

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

SW3

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vtp domain CAMPUS
Changing VTP domain name from NULL to CAMPUS
Switch(config)#vtp mode client
Setting device to VTP CLIENT mode.
Switch(config)#vtp password cisco
Setting device VLAN database password to cisco
Switch(config)#
Switch(config)#end
Switch#write%SW_VLAN-6-VTP_DOMAIN_NAME_CHG: VTP domain name changed to
CAMPUS.

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

Create VLANs

Only create them on the VTP Server (SW2).

SW2

```
[OK]
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name MGMT
Switch(config-vlan)#
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name STAFF
Switch(config-vlan)#
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name STUDENTS
Switch(config-vlan)#
Switch(config-vlan)#vlan 40
Switch(config-vlan)#name GUEST
Switch(config-vlan)#
Switch(config-vlan)#vlan 99
Switch(config-vlan)#name NATIVE
Switch(config-vlan)#
Switch(config-vlan)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

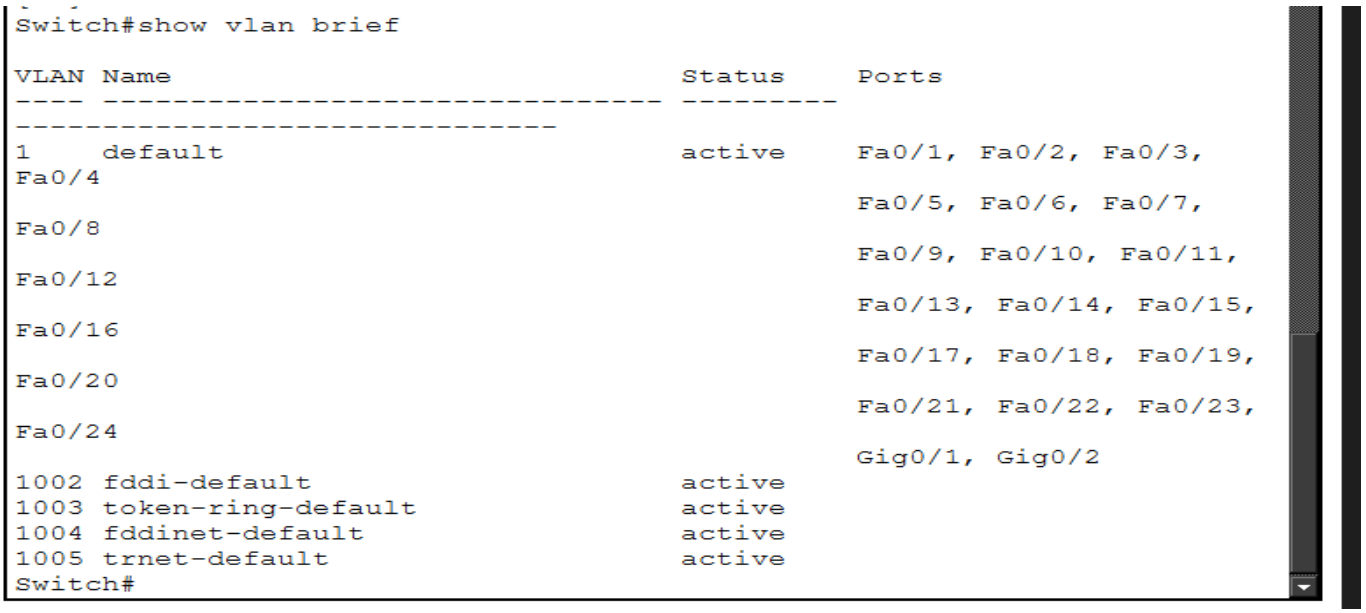
```
Switch#show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- ---------
--------------------------------
1    default                             active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4
                                                   Fa0/5, Fa0/6, Fa0/7,
Fa0/8
                                                   Fa0/9, Fa0/10, Fa0/11,
Fa0/12
                                                   Fa0/13, Fa0/14, Fa0/15,
Fa0/16
                                                   Fa0/17, Fa0/18, Fa0/19,
Fa0/20
                                                   Fa0/21, Fa0/22, Fa0/23,
Fa0/24
                                                   Gig0/1, Gig0/2
10   MGMT                                active
20   STAFF                               active
30   STUDENTS                            active
40   GUEST                               active
99   NATIVE                              active
1002 fddi-default                        active
1003 token-ring-default                  active
1004 fddinet-default                     active
1005 trnet-default                       active
Switch#
```

Also show on SW1 AND SW3

```
Switch#show vlan brief

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4
                                                 Fa0/5, Fa0/6, Fa0/7,
Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11,
Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15,
Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19,
Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23,
Fa0/24
                                                 Gig0/1, Gig0/2
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#
```



Configure Access Ports

SW1

According to the lab:

- Fa0/1-Fa0/10 → VLAN 20
- Fa0/11-Fa0/20 → VLAN 30

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fa0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#
Switch(config-if-range)#interface range fa0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#
Switch(config-if-range)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

SW3

Let's configure:

- Fa0/1-Fa0/10 → VLAN 30
- Fa0/11-Fa0/20 → VLAN 40

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fa0/1-10
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#
Switch(config-if-range)#interface range fa0/11-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 40
Switch(config-if-range)#
Switch(config-if-range)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

Configure Trunks

From your diagram:

- SW1 G0/1 ↔ SW2 G0/1
- SW2 G0/2 ↔ SW3 G0/2

These links must be trunks.

SW1

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#
Switch(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
GigabitEthernet0/1 (99), with Switch GigabitEthernet0/1 (1).

Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

SW2

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface g0/1
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can
not be configured to "trunk" mode.
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#
Switch(config-if)#interface g0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can
not be configured to "trunk" mode.
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

SW3

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface g0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#
Switch(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2,
changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

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Verify

Run on all switches:

SW1

```
Switch#show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- ---------
--------------------------------
1    default                             active    Fa0/21, Fa0/22, Fa0/23,
Fa0/24
                                                   Gig0/2
10   MGMT                                active
20   STAFF                               active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4
                                                   Fa0/5, Fa0/6, Fa0/7,
Fa0/8
                                                   Fa0/9, Fa0/10
30   STUDENTS                            active    Fa0/11, Fa0/12, Fa0/13,
Fa0/14
                                                   Fa0/15, Fa0/16, Fa0/17,
Fa0/18
                                                   Fa0/19, Fa0/20
40   GUEST                               active
99   NATIVE                              active
1002 fddi-default                        active
1003 token-ring-default                  active
1004 fddinet-default                     active
1005 trnet-default                       active
Switch#
```

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig0/1      10,20,30,40,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30,40,99
```

SW2

```
Switch#show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- ---------
--------------------------------
1    default                             active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4
                                                   Fa0/5, Fa0/6, Fa0/7,
Fa0/8
                                                   Fa0/9, Fa0/10, Fa0/11,
Fa0/12
                                                   Fa0/13, Fa0/14, Fa0/15,
Fa0/16
                                                   Fa0/17, Fa0/18, Fa0/19,
Fa0/20
                                                   Fa0/21, Fa0/22, Fa0/23,
Fa0/24
10   MGMT                                active
20   STAFF                               active
30   STUDENTS                            active
40   GUEST                               active
99   NATIVE                              active
1002 fddi-default                        active
1003 token-ring-default                  active
1004 fddinet-default                     active
1005 trnet-default                       active
Switch#
```

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```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      auto         n-802.1q       trunking      99
Gig0/2      auto         n-802.1q       trunking      99

Port        Vlans allowed on trunk
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Switch#
```

SW3

```
Switch#show vlan brief

VLAN Name                                Status    Ports
---- ----------------------------------- --------- -------------------------------
1    default                             active    Fa0/21, Fa0/22, Fa0/23,
Fa0/24
                                                   Gig0/1
10   MGMT                                active
20   STAFF                               active
30   STUDENTS                            active    Fa0/1, Fa0/2, Fa0/3,
Fa0/4
                                                   Fa0/5, Fa0/6, Fa0/7,
Fa0/8
                                                   Fa0/9, Fa0/10
40   GUEST                               active    Fa0/11, Fa0/12, Fa0/13,
Fa0/14
                                                   Fa0/15, Fa0/16, Fa0/17,
Fa0/18
                                                   Fa0/19, Fa0/20
99   NATIVE                              active
1002 fddi-default                        active
1003 token-ring-default                  active
1004 fddinet-default                     active
1005 trnet-default                       active
Switch#
```

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/2      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig0/2      10,20,30,40,99

Port        Vlans allowed and active in management domain
Gig0/2      10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/2      10,20,30,40,99
```

Inter-VLAN Routing Option 1 (Separate Physical Interfaces)

Configure Router R1

On R1:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface fa0/1
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#end
Router#write
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router#
```

Verify

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/0        10.10.10.1      YES manual up
up
FastEthernet0/1        10.10.20.1      YES manual up
up
Vlan1                  unassigned      YES unset  administratively down
down
Router#
```

Configure SW2 Ports Connected to Router

The router interfaces must connect to access ports.

Configure SW2:

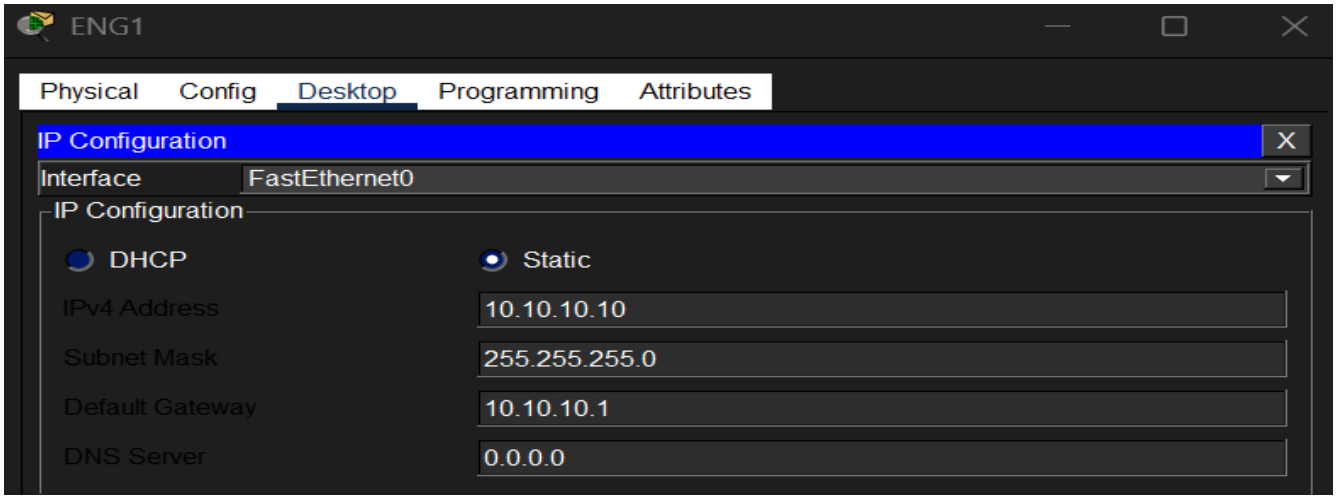
```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Switch#
```

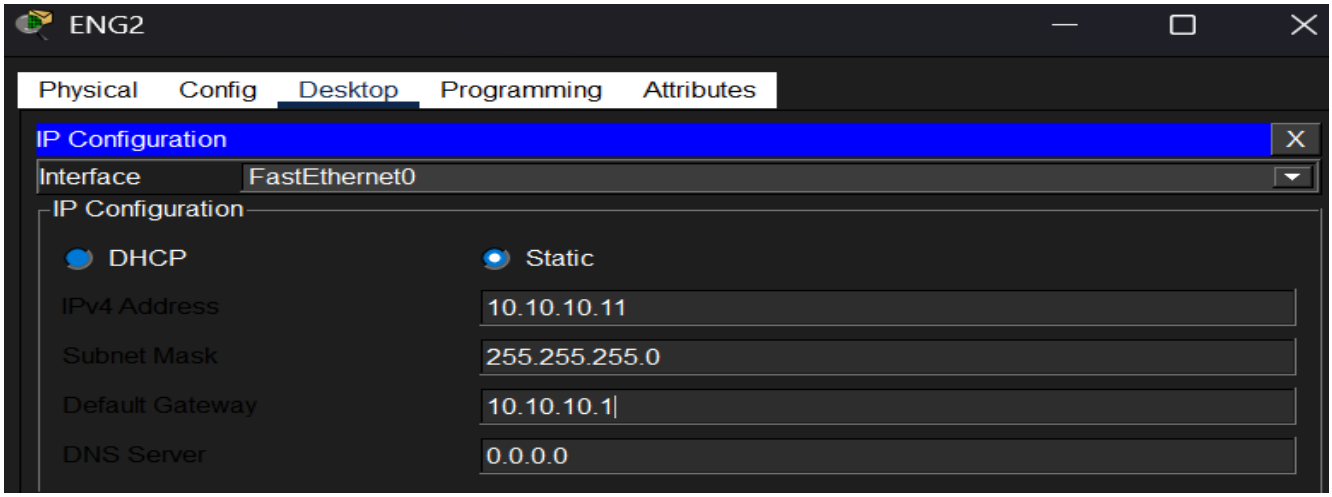
Configure the PCs

Engineering PCs (VLAN 10)

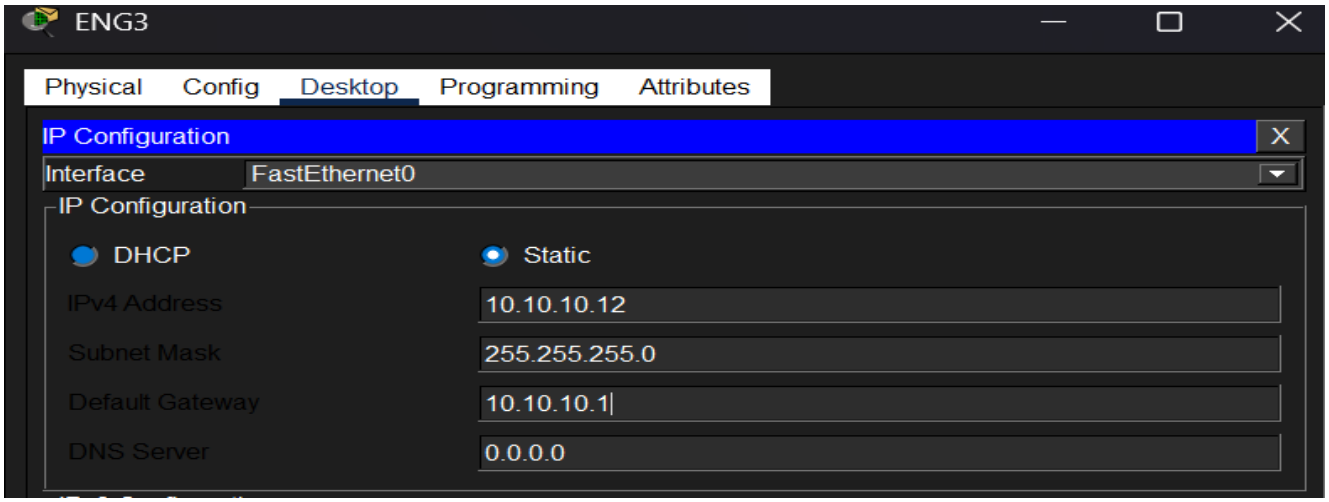
ENG1



ENG2

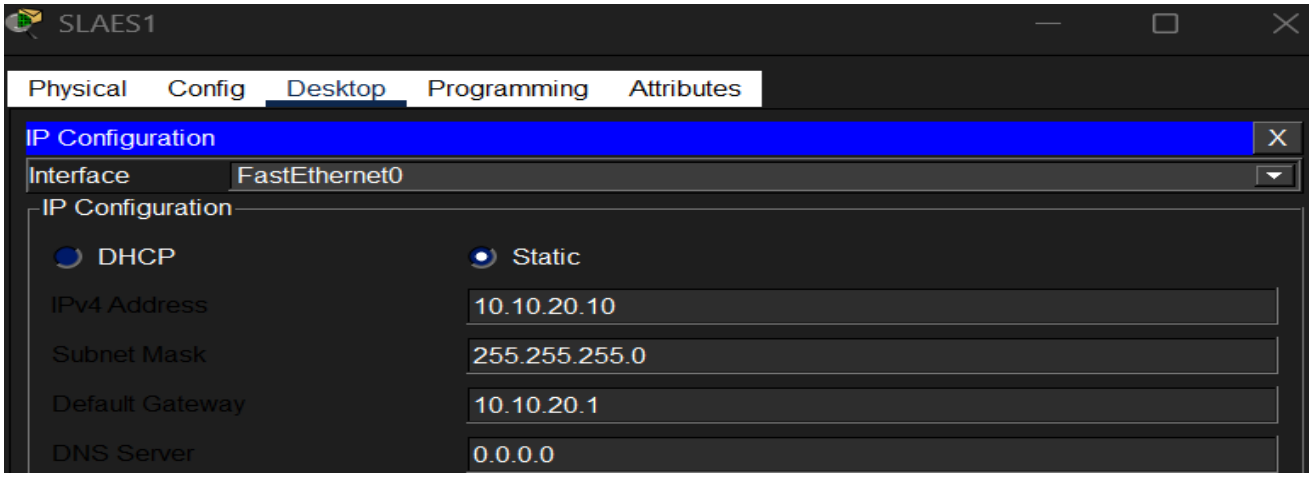


ENG3

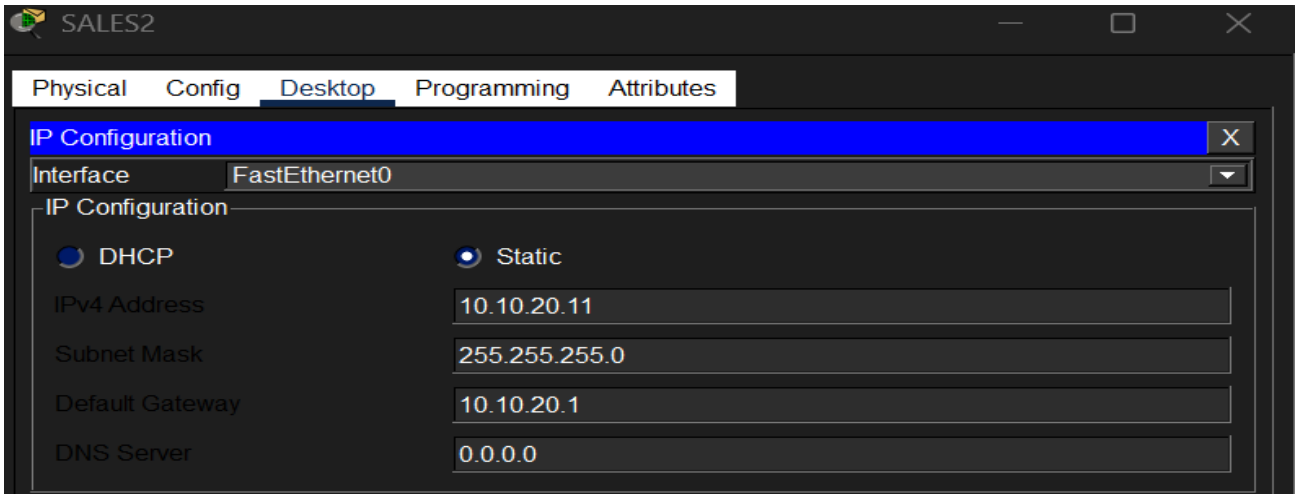


Sales PCs (VLAN 20)

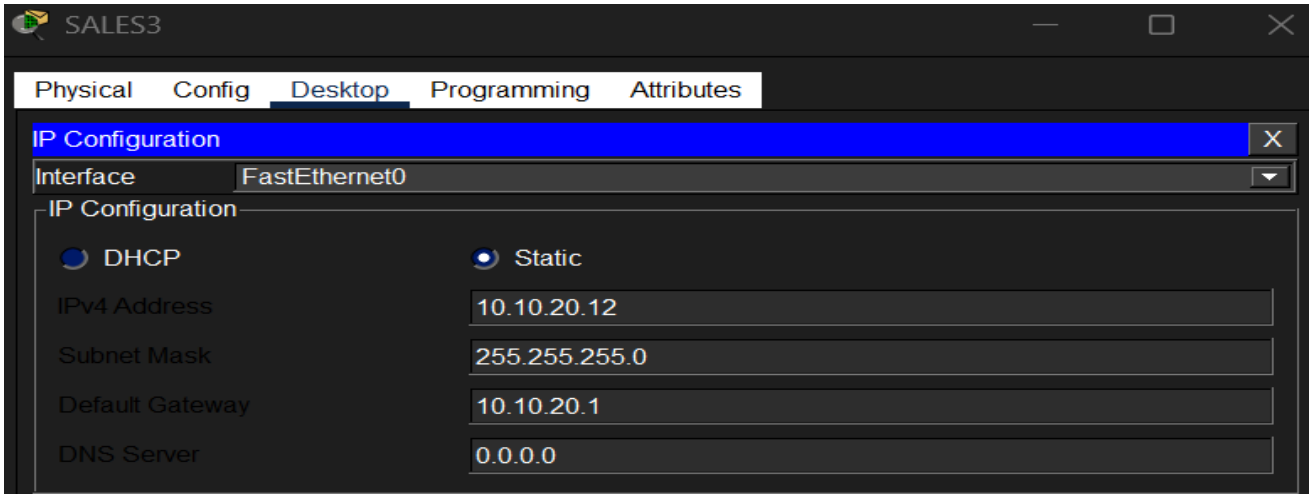
SALES1



SALES2



SLAES3



Verify VLAN Membership

Based on the diagram:

SW1

```
Switch>EN
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#
Switch(config-if-range)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

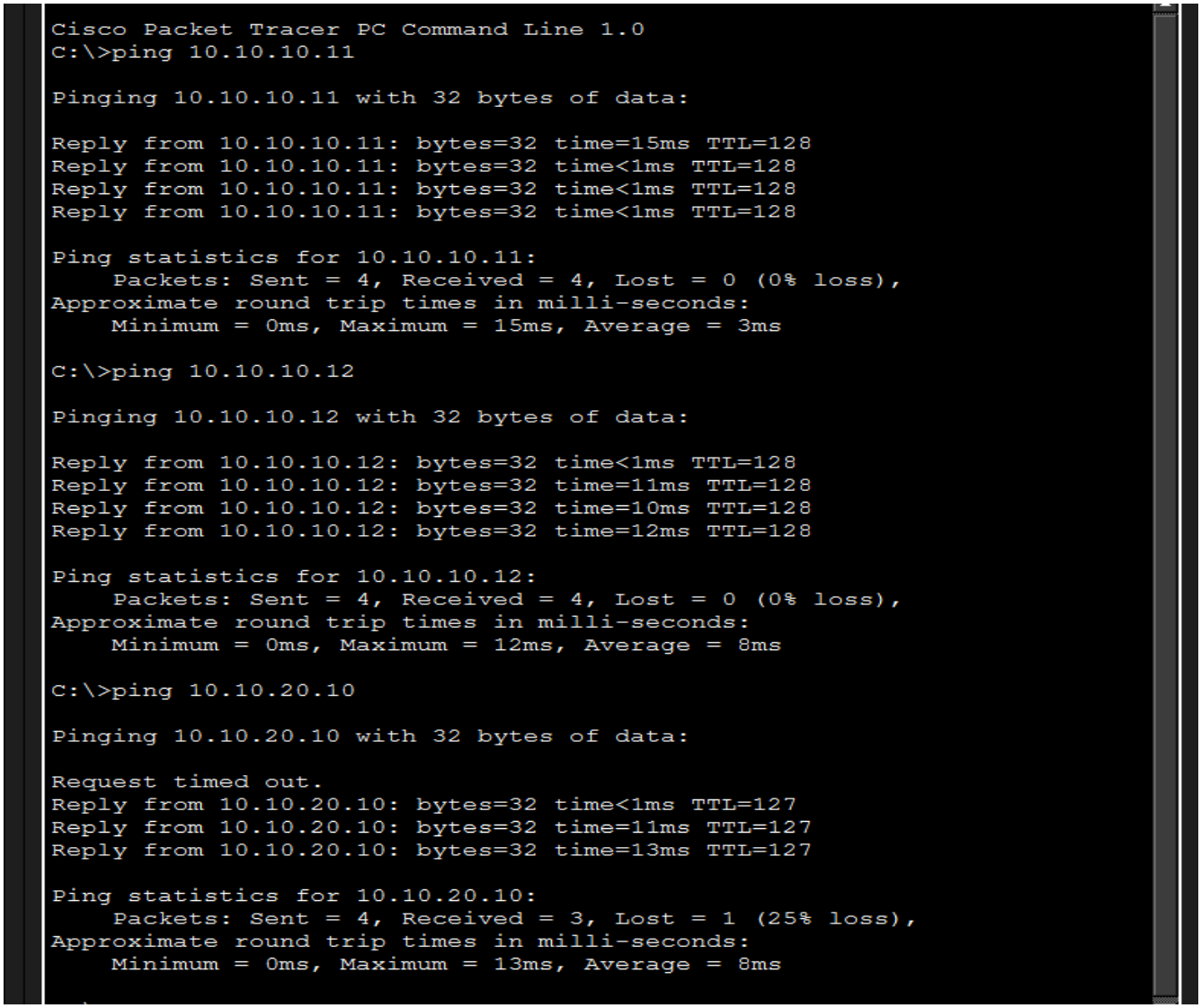
SW3

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#interface fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
Switch(config-if)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

Test Connectivity

From ENG1:


Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.10.11
Pinging 10.10.10.11 with 32 bytes of data:
Reply from 10.10.10.11: bytes=32 time=15ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Reply from 10.10.10.11: bytes=32 time<1ms TTL=128
Ping statistics for 10.10.10.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 3ms
C:\>ping 10.10.10.12
Pinging 10.10.10.12 with 32 bytes of data:
Reply from 10.10.10.12: bytes=32 time<1ms TTL=128
Reply from 10.10.10.12: bytes=32 time=11ms TTL=128
Reply from 10.10.10.12: bytes=32 time=10ms TTL=128
Reply from 10.10.10.12: bytes=32 time=12ms TTL=128
Ping statistics for 10.10.10.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 12ms, Average = 8ms
C:\>ping 10.10.20.10
Pinging 10.10.20.10 with 32 bytes of data:
Request timed out.
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time=11ms TTL=127
Reply from 10.10.20.10: bytes=32 time=13ms TTL=127
Ping statistics for 10.10.20.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 13ms, Average = 8ms

```
C:\>ping 10.10.20.11

Pinging 10.10.20.11 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.11: bytes=32 time=12ms TTL=127
Reply from 10.10.20.11: bytes=32 time<1ms TTL=127
Reply from 10.10.20.11: bytes=32 time=12ms TTL=127

Ping statistics for 10.10.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 8ms

C:\>ping 10.10.20.12

Pinging 10.10.20.12 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.12: bytes=32 time=11ms TTL=127
Reply from 10.10.20.12: bytes=32 time=11ms TTL=127
Reply from 10.10.20.12: bytes=32 time=12ms TTL=127

Ping statistics for 10.10.20.12:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 12ms, Average = 11ms
```

Inter-VLAN Routing Option 2 (Router-on-a-Stick)

This method uses **one physical router interface** and **sub interfaces**.

Step 1: Remove Option 1 Configuration

On R1:

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#no ip address
Router(config-if)#shutdown

Router(config-if)#
Router(config-if)#interface fa0/1
Router(config-if)#no ip address
Router(config-if)#shutdown

Router(config-if)#
Router(config-if)#end
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to down

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to
administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed
state to down

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Step 2: Use One Link

Keep only:
SW2 Fa0/1 <--------> R1 Fa0/0
Remove the cable between:
SW2 Fa0/2 <--------> R1 Fa0/1

Step 3: Configure SW2 Port as a Trunk

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#
Switch(config-if)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

Verify:

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      auto         n-802.1q       trunking      99
Gig0/2      auto         n-802.1q       trunking      99

Port        Vlans allowed on trunk
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Port        Vlans allowed and active in management domain
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20,30,40,99
Gig0/2      10,20,30,40,99
```

Step 4: Configure Router Subinterfaces

On R1:

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface fa0/0
Router(config-if)#no shutdown

Router(config-if)#
Router(config-if)#interface fa0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 10.10.10.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface fa0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 10.10.20.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface fa0/0.30
Router(config-subif)#encapsulation dot1Q 30
Router(config-subif)#ip address 10.10.30.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface fa0/0.40
Router(config-subif)#encapsulation dot1Q 40
Router(config-subif)#ip address 10.10.40.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#interface fa0/0.99
Router(config-subif)#encapsulation dot1Q 99 native
Router(config-subif)#
Router(config-subif)#end
Router#write
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed
state to up

%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10,
changed state to up

%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20,
```

Step 5: Verify

```
Router#show ip interface brief
Interface              IP-Address      OK? Method Status
Protocol
FastEthernet0/0        unassigned      YES manual up
up
FastEthernet0/0.10     10.10.10.1      YES manual up
up
FastEthernet0/0.20     10.10.20.1      YES manual up
up
FastEthernet0/0.30     10.10.30.1      YES manual up
up
FastEthernet0/0.40     10.10.40.1      YES manual up
up
FastEthernet0/0.99     unassigned      YES unset  up
up
FastEthernet0/1        unassigned      YES manual administratively down
down
Vlan1                  unassigned      YES unset  administratively down
down
Router#
```

Step 6: Test Connectivity

From ENG1:

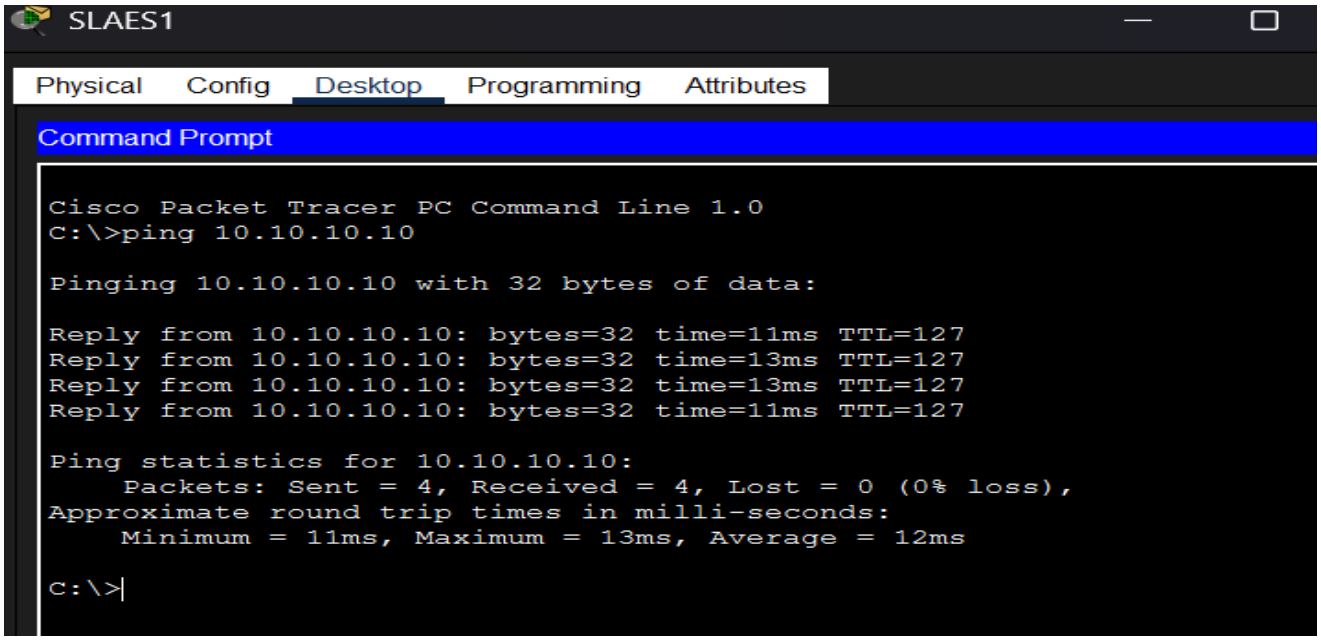
```
C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

From SALES1:



Task 7: Layer 3 Switch Inter-VLAN Routing

Step 1: Disconnect the Router

For this task, the instructions say:

"verify inter-VLAN connectivity without using the router at all"

Remove the connection between:

R1 → SW2

Step 2: Enable Routing on SW2

On SW2:

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#ip routing
Switch(config)#
Switch(config)#end
Switch#write
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

Step 3: Create VLAN Interfaces (SVIs)

On SW2:

```
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#interface vlan 10
Switch(config-if)#ip address 10.10.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 20
Switch(config-if)#ip address 10.10.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 30
Switch(config-if)#ip address 10.10.30.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#interface vlan 40
Switch(config-if)#ip address 10.10.40.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#
Switch(config-if)#end
Switch#write
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to
up

%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to
up

%LINK-5-CHANGED: Interface Vlan30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to
up

%LINK-5-CHANGED: Interface Vlan40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to
up
```

Step 4: Verify SVIs

```
down
GigabitEthernet0/1     unassigned      YES unset  up
up
GigabitEthernet0/2     unassigned      YES unset  up
up
Vlan1                  unassigned      YES unset  administratively down
down
Vlan10                 10.10.10.1      YES manual up
up
Vlan20                 10.10.20.1      YES manual up
up
Vlan30                 10.10.30.1      YES manual up
up
Vlan40                 10.10.40.1      YES manual up
up
Switch#
```

Verify Routing Table

On SW2:

```
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 4 subnets
C       10.10.10.0 is directly connected, Vlan10
C       10.10.20.0 is directly connected, Vlan20
C       10.10.30.0 is directly connected, Vlan30
C       10.10.40.0 is directly connected, Vlan40
```

Reflection Question 8

Of the three inter-VLAN routing methods you configured, which would scale best in a large campus network, and why?

Answer:

The **Layer 3 Switch method** scales best in a large campus network. Unlike using separate router interfaces or Router-on-a-Stick, a Layer 3 switch performs routing directly in hardware at wire speed, resulting in higher performance and lower latency. It also eliminates the single trunk bottleneck between a router and switch, supports many VLANs more efficiently, and is easier to expand as the network grows. For these reasons, multilayer switches are commonly used in enterprise and campus network environments for inter-VLAN routing.